Role	*Must, Should, Could, Wants (PRRIORITY ORDER)*	*Action*	*Hardware or Software?*
User	Should	See alerts for pH regulation, % nutrient change	S
User	Must	See a list of templated options (apples, strawberries, etc.) to select from	S
User	Must	Be able to modify individual parameters for the fertilizer.	S
User	Must	Be able to select pre-created set of parameters for a specific fruit/vegetable.	S
User	Wants	To be able to share a set of saved parameters with other app users.	S
User	Wants	To be able to communicate with other farmers on the application.	S
User	Must	Be able to favourite a specific fertilizer setting so that they can save it for future use or so they can log it.	S
User	Should	Be able to fine tune their settings to account for specific regions or countries.	S
User	Must	be able to select a specific fruit/vegetable setting and fine tune the pre created setting even further.	S
User	Wants	To be able to order duckweed from the app.	S
User	Must	Be able to see the time until fruit/vegetable growth.	S
User	Must	be able to monitor the temperature of water, pH, CO2 concentration at the water's surface, light, duckweed biomass, nutrient concentration in the water, and the airation of the water.	S

User	Should	Be able to receive alerts when parameters are out of ideal ranges.	S
User	Must	Be able to log a description for a saved setting.	S
User	Could	Be able to view historical performance metrics for a specific set of parameters.	S
User	Should	Be able to receive alerts on overgrowth of opportunistic microorganisms which may impede crop yield.	S

- Viewing nutrients of the item
    1. ?
- Farmer wants to grow potatoes; the app tunes the fertilizer to what the farmer is trying to grow
    - If you press a specific fruit, the app will regulate the duckweed to support the optimal environment for that fruit or vegetable
    - User must be able to tune the fertilizer to a specific fruit/vegetable they are trying to grow
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- Manual mode as well would be cool
    - Essentially be able to modify specific parameters as well
    - User must be able to modify specific parameters for the fertilizer
- App opens, theres a dashboard page
        - Manual mode – given nitrogen % count, etc. to custom-tune
    - User must be able to select pre-created parameters for a specific fruit/vegetable.
- Shareable customizable options (WANT)
    - User would want to be able to share a set of saved parameters with other farmers.
- Platform for farmers to communicate
    - User would want to be able to communicate with other farmers on the application.
- Heart for a favourite setting that is easily shareable

    - User must be able to favourite a specific fertilizer setting so that they can save it for future use or so they can log it.
- Customize the settings for their specific regions and countries
    - User must be able to fine tune their settings to account for specific regions or countries.
- Temperature control
- Consider biomass density
- Want it to grow optimally: set it so it grows in a x number of days
- X variables: you have potato mode but you can even customize the potato mode a bit
    - User must be able to select a specific fruit/vegetable setting and fine tune the pre created setting even further.
- Like uber eats, you can order your duckweed from the app
    - User wants to be able to order duckweed from the application.
- Non-manual options:
    - Fruit
    - Temperature
    -
- See time until growth
    - User must be able to see the time until fruit/vegetable growth.
- What parameters do we want monitored
    - Temperature of water, pH (CO2 concentration can be reflected here, one in the water and one in the surface of the water for CO2 concentration), light, duckweed biomass, Nutrient concentration in water, airation/turbiduty (dissolved content of the water to be measured),
    - Ideally coculturing bacteria with cyanobacteria
    - User must be able to monitor the temperature of water, pH, CO2 concentration at the water's surface, light, duckweed biomass, nutrient concentration in the water, and the airation of the water.
- Alerts for when certain things happen in the cultivator
    - User should be able to receive alerts when parameters are out of ideal ranges.
- When you have a setting you save, would be cool if we can have a description
    - User Must be able to log a description for a saved setting.
- Main focus as of right now should not have the sharing option, but having the bookmark with description for a setting will be cool.
- Performance metrics; crop efficiency

    - How a specific history or setting affects the performance (even comparing two)
    - User Could be able to view historical performance metrics for a specific set of parameters.
    - Concern with performance metrics; how can a person like record their final yields?
- Administrator version; engineering controls for
    - User could be able to
- Anything to measure opportunistic microorganism (contaminations).
    - Any sensors that can be implemented to say we have x levels of this metabolite, indicating the presence of this
    - User should be able to receive alerts on overgrowth of opportunistic microorganisms which may impede crop yield.

- For mindfueld: assume wastewater

HARDWARE

Sealed bag for dessication

Any sensors that can be implemented to say we have x levels of this metabolite, indicating the presence of this

Mind fueld

- Proof of concept
- Figma high fidelity prototype

User Stories

As a…	I want to…	So that…
User	See digital alerts for pH regulation, % nutrient change, and other outlier variables	I can be told when something is going haywire or wrong in my cultivator
User	See a list of templated options (apples, strawberries, etc.) to select from	I have templated nutrient profiles derived from public sources to grow my duckweed
User	Be able to modify individual parameters for the fertilizer inside of a Manual Mode	Individual parameters of the cultivator's properties (that are pre-determined by the templates) can be tuned to my preferences
Engineer & User	Be able to have my own list of 'Engineer' properties on the cultivator	I can change important hardware properties inside of the cultivator (such as water depth) through the dashboard
Engineer & User	Be able to turn off the application on the dashboard	The application turns off when I want to (duh)
Engineer	Have the cultivator turn off when the Raspberry Pi or cultivator exceeds a certain temperature	It does not go on fire
User	Be able to favourite a specific fertilizer setting	I can save my settings/preferences for future reference
User	Upload a fertilizer setting via a designated file	Receive and/or upload settings/preferences for the cultivator
User	Fine-tune my settings to account for my region	The cultivator can work effectively in different climates
User	Order duckweed from the app	I have duckweed from the app
User	Be able to see the time until fruit/vegetable growth	I know when it is time to harvest the duckweed through the cultivator
User	Be able to harvest the duckweed through the press of a button	It does not need to be done manually and can be automated through the app
User	Be able to monitor the temperature of water, pH, CO2 concentration at the water's surface, light, duckweed biomass, nutrient concentration in the water, and the aeration of the water.	I can view the analytics of the duckweed in a dashboard to understand its health

User	Be able to log a description for my saved preferences	Be able to log a description for a saved setting.
User	Could	Be able to view historical performance metrics for a specific set of parameters.
User	Be able to receive alerts on overgrowth of opportunistic microorganisms which may impede crop yield	I know when overgrowth may be ruining my cultivator
Engineer	Seek automatic shutdown of the cultivator when parameters exceed typical values	The cultivator does not break when certain parameters go too far
Engineer	Have the Raspberry Pi restart every night	It does not behave improperly when running 24/7
Arduino	Turn off the machine when its too hot	The machine does not overheat
Arduino	Edit the lights dimmer properties	Edit the temperature of the lights onto the duckweed
Arduino	Use API Calls onto a wi-fi stick	Send the data to a database for interface retrieval
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Arduino		

- Using a RTC (real time clock) to monitor date, time and temperature of the system
- A wifi stick to connect to the interface
- MOSFITS as a dimming switch for the lights
- LED lights connected to breakout board, connected to the arduino
- Lights should be able to:
    - Dim/brighten

- Four lights:
    - Red, blue LEDs
    - Far red lights
    - UVB sensitive lights
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